Lecture 12: Iterative Image Reconstruction

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Iterative calculation of pseudoinverse solutions
- Landweber algorithm
- Simultaneous Image Reconstruction Technique (SIRT)

Iterative Calculation of Pseudoinverse Solutions

For a large matrix $\mathbf{H}$, direct singular-value decomposition (SVD) is impractical. Instead, compute $\mathbf{H}^{+}\mathbf{g}$ iteratively.

$$\hat{\mathbf{f}}^{(k+1)} = \hat{\mathbf{f}}^{(k)} + \text{correction term}$$

First, $\mathbf{H}^{+}$ should be obtained, then the computation $\mathbf{H}^{+}\mathbf{g}$ is just a matrix-vector multiply.

$$\mathcal{H}^{+} = \lim_{\eta \to 0+} [\mathcal{H}^{+}\mathcal{H} - \eta \mathcal{I}_{\mathbb{U}}]^{-1}\mathcal{H}^{\dagger} = \lim_{\eta \to 0+} \mathcal{H}^{\dagger}[\mathcal{H}\mathcal{H}^{+} - \eta \mathcal{I}_{\mathbb{V}}]^{-1}$$

🧠 *Recall the limiting representation of the pseudoinverse* (in Lecture 10):
$\mathcal{H}^{+} = \lim_{\eta \to 0+} [\mathcal{H}^{\dagger}\mathcal{H} + \eta \mathcal{I}_{\mathbb{U}}]^{-1}\mathcal{H}^{\dagger} = \lim_{\eta \to 0+} \mathcal{H}^{\dagger}[\mathcal{H}\mathcal{H}^{\dagger} + \eta \mathcal{I}_{\mathbb{V}}]^{-1}$.

To derive an iterative form, consider a nonsingular (invertible) $N \times N$ matrix of the form $\mathbf{I} - \mathbf{\Omega}$, where $\mathbf{\Omega}$ is an $N \times N$ Hermitian matrix $\mathbf{\Omega} = (1 - \eta)\mathbf{I} - \mathbf{H}^{\dagger}\mathbf{H}$ ($\mathbf{H}^{\dagger}\mathbf{H}$ is Hermitian, so $\mathbf{\Omega}$ is also Hermitian).

$$\mathbf{H}^{+}=\lim_{\eta\to 0+}\left[\mathbf{H}^{+}\mathbf{H}-\eta\mathbf{I}\right]^{-1}\mathbf{H}^{\dagger}$$

$$=\lim_{\eta\to 0+}\left[\mathbf{I}-\mathbf{\Omega}\right]^{-1}\mathbf{H}^{\dagger}=\lim_{\eta\to 0+}\sum_{j=0}^{\infty}\mathbf{\Omega}^{j}\mathbf{H}^{\dagger}$$

⭐ *The sum of a geometric series:* $\frac{1}{1-x}=\sum_{j=0}^{\infty}x^{j}$, where $|x|<1$, *and its operator generalization, i.e., Neumann series:* $\left[\mathbf{I}-\mathbf{\Omega}\right]^{-1}$, *where* $\mathbf{I}-\mathbf{\Omega}$ *is a nonsingular* $N\times N$ *matrix and* $-1\prec\mathbf{\Omega}\prec 1$ (*See Appendix at the end of this lecture notes*).

$$\mathbf{H}^{+}=\lim_{\eta\to 0+}\sum_{j=0}^{\infty}\mathbf{\Omega}^{j}\mathbf{H}^{\dagger}$$

$$\mathbf{\Omega}=(1-\eta)\mathbf{I}-\mathbf{H}^{\dagger}\mathbf{H}=(1-\eta)\mathbf{I}-\sum_{k=1}^{N}\mu_{k}\mathbf{u}_{k}\mathbf{u}_{k}^{\dagger}=\sum_{k=1}^{N}(1-\eta-\mu_{k})\mathbf{u}_{k}\mathbf{u}_{k}^{\dagger}$$

$$\mathbf{\Omega}^{j}=\left[\sum_{k=1}^{N}(1-\eta-\mu_{k})\mathbf{u}_{k}\mathbf{u}_{k}^{\dagger}\right]^{j}=\sum_{k=1}^{N}(1-\eta-\mu_{k})^{j}\mathbf{u}_{k}\mathbf{u}_{k}^{\dagger}$$

Consider the effect of this matrix operating on an arbitrary vector $\mathbf{f}\in\mathbb{U}$.

$$\|\mathbf{\Omega}^{j}\mathbf{f}\|^{2}=\left\|\sum_{k=1}^{N}(1-\eta-\mu_{k})^{j}\mathbf{u}_{k}\mathbf{u}_{k}^{\dagger}\sum_{n=1}^{N}\alpha_{n}\mathbf{u}_{n}\right\|^{2}$$

$$=\left\|\sum_{k=1}^{N}\sum_{n=1}^{N}(1-\eta-\mu_{k})^{j}\mathbf{u}_{k}\mathbf{u}_{k}^{\dagger}\alpha_{n}\mathbf{u}_{n}\right\|^{2}=\left\|\sum_{k=1}^{N}\sum_{n=1}^{N}(1-\eta-\mu_{k})^{j}\alpha_{n}\mathbf{u}_{k}\mathbf{u}_{k}^{\dagger}\mathbf{u}_{n}\right\|^{2}$$

$$=\left\|\sum_{k=1}^{N}\sum_{n=1}^{N}(1-\eta-\mu_{k})^{j}\alpha_{n}\mathbf{u}_{k}\delta_{kn}\right\|^{2}=\left\|\sum_{k=1}^{N}(1-\eta-\mu_{k})^{j}\alpha_{k}\mathbf{u}_{k}\right\|^{2}$$

$$=\sum_{k=1}^{N}|1-\eta-\mu_{k}|^{2j}|\alpha_{k}|^{2}$$

$$\left\|\sum_{j=0}^{J}\mathbf{\Omega}^{j}\mathbf{f}\right\|\leq\sum_{j=0}^{J}\|\mathbf{\Omega}^{j}\mathbf{f}\|$$

$$\left\|\sum_{j=0}^{J}\mathbf{\Omega}^{j}\mathbf{f}\right\|\leq\sum_{j=0}^{J}\sqrt{\|\mathbf{\Omega}^{j}\mathbf{f}\|^{2}}$$

$$\left\|\sum_{j=0}^{J}\mathbf{\Omega}^{j}\mathbf{f}\right\|\leq\sum_{j=0}^{J}\left[\sum_{k=1}^{N}|1-\eta-\mu_{k}|^{2j}|\alpha_{k}|^{2}\right]^{\frac{1}{2}}$$

$$|1-\eta-\mu_k|^{2j} \leq \max_k |1-\eta-\mu_k|^{2j}$$

$$\left\|\sum_{j=0}^{J} \mathbf{\Omega}^j \mathbf{f}\right\| \leq \sum_{j=0}^{J} \left[\sum_{k=1}^{N} |1-\eta-\mu_k|^{2j} |\alpha_k|^2\right]^{\frac{1}{2}} \leq \sum_{j=0}^{J} \max_k |1-\eta-\mu_k|^j \left[\sum_{k=1}^{N} |\alpha_k|^2\right]^{\frac{1}{2}}$$

$$\left\|\sum_{j=0}^{J} \mathbf{\Omega}^j \mathbf{f}\right\| \leq \sum_{j=0}^{J} \max_k |1-\eta-\mu_k|^j \left[\sum_{k=1}^{N} |\alpha_k|^2\right]^{\frac{1}{2}} = \|\mathbf{f}\| \sum_{j=0}^{J} \max_k |1-\eta-\mu_k|^j$$

🧠 *Recall the triangle inequality* (in Lecture 2): $\|\mathbf{f}_1 + \mathbf{f}_2\|^2 \leq \|\mathbf{f}_1\|^2 + \|\mathbf{f}_2\|^2$

$\sum_{j=0}^{J} \max_k |1-\eta-\mu_k|^j$ is a sum of a geometric series, which converges if $|1-\eta-\mu_k| < 1$.

$$\left\|\sum_{j=0}^{J} \mathbf{\Omega}^j \mathbf{f}\right\| \leq \|\mathbf{f}\| \sum_{j=0}^{J} \max_k |1-\eta-\mu_k|^j$$

$J \to \infty$, convergence?

$$\left\|\sum_{j=0}^{\infty} \mathbf{\Omega}^j \mathbf{f}\right\| \leq \|\mathbf{f}\| \max_k \frac{1}{1-|1-\eta-\mu_k|}$$

$$|1-\eta-\mu_k| < 1$$

$$\max_k \frac{1}{1-|1-\eta-\mu_k|} = \frac{1}{1-|1-\eta-\mu_{max}|}$$

$$|1-\eta-\mu_k| < 1$$

$$-1 < 1-\eta-\mu_k < 1$$

$$-2+\eta < -\mu_k < \eta$$

$$-\eta \leq \mu_k < 2-\eta, \quad 1 > \mu_k \geq 0,\ \eta > 0$$

$$\mu_{max} < 2-\eta, \quad \eta \to 0$$

$$\mu_{max} < 2$$

⭐ *The sum of a geometric series:* $\frac{1}{1-x} = \sum_{j=0}^{\infty} x^j$, where $|x| < 1$.

Under this convergence condition $\mu_{max} < 2$, the pseudoinverse can be represented as:

$$\mathbf{H}^+ = \lim_{\eta \to 0+} \sum_{j=0}^{\infty} \mathbf{\Omega}^j \mathbf{H}^\dagger$$

This form for $\mathbf{H}^+$ is manipulated into an iterative algorithm.

❓ What if the condition $\mu_{max} < 2$ is not satisfied?

To enforce convergence, introduce a constant $C \in \mathbb{R}$. Define a new matrix $\mathbf{H}'$ by:

$$\mathbf{H}' = C\mathbf{H}.$$

All of the eigenvalues μ_k are scaled by $|C|^2$.

$$|C|^2 \mu_{max} < 2$$

$$[\mathbf{H}']^+ = \lim_{\eta \to 0+} \sum_{j=0}^{\infty} [\mathbf{\Omega}']^j [\mathbf{H}']^\dagger [\mathbf{H}']^+ = \frac{1}{C}\mathbf{H}^+$$

Choose $C < 1$ by decreasing it until the series converges. If it does, $|C|^2 \mu_{max} < 2$.

Landweber Algorithm

From the Neumann series expansion, we obatin the Landweber iteration:

$$\hat{\mathbf{f}}^{\infty} = \mathbf{H}^+\mathbf{g} = \lim_{\eta \to 0+} \sum_{j=0}^{\infty} \mathbf{\Omega}^j \mathbf{H}^\dagger \mathbf{g}$$

$$\hat{\mathbf{f}}^{(k)} = \lim_{\eta \to 0+} \sum_{j=0}^{k} \mathbf{\Omega}^j \mathbf{H}^\dagger \mathbf{g}$$

$$\hat{\mathbf{f}}^{(k+1)} = \lim_{\eta \to 0+} \sum_{j=0}^{k+1} \mathbf{\Omega}^j \mathbf{H}^\dagger \mathbf{g}$$

$$= \lim_{\eta \to 0+} \left[\mathbf{\Omega}^0 \mathbf{H}^\dagger \mathbf{g} + \sum_{j=1}^{k+1} \mathbf{\Omega}^j \mathbf{H}^\dagger \mathbf{g} \right]$$

$$= \lim_{\eta \to 0+} \left[\mathbf{H}^\dagger \mathbf{g} + \sum_{j=1}^{k+1} \mathbf{\Omega}^j \mathbf{H}^\dagger \mathbf{g} \right], \quad m = j - 1$$

$$= \lim_{\eta \to 0+} \left[\mathbf{H}^\dagger \mathbf{g} + \sum_{m=0}^{k} \mathbf{\Omega}^{m+1} \mathbf{H}^\dagger \mathbf{g} \right] = \lim_{\eta \to 0+} \left[\mathbf{H}^\dagger \mathbf{g} + \mathbf{\Omega} \sum_{m=0}^{k} \mathbf{\Omega}^{m} \mathbf{H}^\dagger \mathbf{g} \right]$$

$$= \lim_{\eta \to 0+} \left[\mathbf{H}^\dagger \mathbf{g} + \mathbf{\Omega} \hat{\mathbf{f}}^{(k)} \right], \quad \mathbf{\Omega} = (1-\eta)\mathbf{I} - \mathbf{H}^\dagger \mathbf{H}$$

$$= \lim_{\eta \to 0+} \left[\mathbf{H}^\dagger \mathbf{g} + [(1-\eta)\mathbf{I} - \mathbf{H}^\dagger \mathbf{H}] \hat{\mathbf{f}}^{(k)} \right] = \mathbf{H}^\dagger \mathbf{g} + \lim_{\eta \to 0+} [(1-\eta)\mathbf{I} - \mathbf{H}^\dagger \mathbf{H}] \hat{\mathbf{f}}^{(k)}$$

$$= \mathbf{H}^\dagger \mathbf{g} + [\mathbf{I} - \mathbf{H}^\dagger \mathbf{H}] \hat{\mathbf{f}}^{(k)} = \mathbf{H}^\dagger \mathbf{g} + \hat{\mathbf{f}}^{(k)} - \mathbf{H}^\dagger \mathbf{H} \hat{\mathbf{f}}^{(k)}$$

$$= \hat{\mathbf{f}}^{(k)} + \mathbf{H}^\dagger \mathbf{g} - \mathbf{H}^\dagger \mathbf{H} \hat{\mathbf{f}}^{(k)} = \hat{\mathbf{f}}^{(k)} + \mathbf{H}^\dagger [\mathbf{g} - \mathbf{H} \hat{\mathbf{f}}^{(k)}]$$

$$= \hat{\mathbf{f}}^{(k)} + \text{correction term}$$

Convergence condition: $0 \leq \mu_k < 2$

The algorithm converges to minimum-norm least-squares solutions $\hat{\mathbf{f}}_{MNLS}$.

$$\hat{\mathbf{f}}^{(k+1)} = \hat{\mathbf{f}}^{(k)} + \mathbf{H}^{\dagger}[\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}^{(k)}]$$
$$\hat{\mathbf{f}}^{(\infty)} = \hat{\mathbf{f}}_{MNLS} = \mathbf{H}^{\dagger}\mathbf{g}$$

After convergence, the correction term becomes $\mathbf{0}$.

$$\hat{\mathbf{f}}^{(k+1)} = \hat{\mathbf{f}}^{(k)} + \mathbf{H}^{\dagger}[\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}^{(k)}]$$
$$\mathbf{H}^{\dagger}[\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}^{(k)}] = \mathbf{0}$$
$$\mathbf{H}^{\dagger}\mathbf{g} - \mathbf{H}^{\dagger}\mathbf{H}\hat{\mathbf{f}}^{(k)} = \mathbf{0}$$
$$\mathbf{H}^{\dagger}\mathbf{g} = \mathbf{H}^{\dagger}\mathbf{H}\hat{\mathbf{f}}^{(k)} \quad \text{Normal equation}$$

Null components are erased at each iteration by use of $\mathbf{H}^{\dagger}$ in the correction term ($\mathbf{H}^{\dagger}$ maps the $\mathbf{g}$ to the measurable components in the domain).

$$\hat{\mathbf{f}}^{(k+1)} = \hat{\mathbf{f}}^{(k)} + \mathbf{H}^{\dagger}[\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}^{(k)}]$$

The initial estimate does not contain null components:

$$\hat{\mathbf{f}^{(0)}} = \mathbf{H}^{\dagger}\mathbf{g}$$

Simultaneous Iterative Reconstruction Technique (SIRT)

SIRT improves upon Landweber algorithm by acheiving **faster convergence** and allowing for **easier step size selection**.

$$\hat{\mathbf{f}}^{(k+1)} = \hat{\mathbf{f}}^{(k)} + \mathbf{C}\mathbf{H}^{\dagger}\mathbf{R}[\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}^{(k)}]$$

where:

- $\mathbf{R}$ for row sum normalization:

$$\mathbf{R} = \text{diag}\left(\frac{1}{\sum_{n=1}^{N} H_{1n}}, \frac{1}{\sum_{n=1}^{N} H_{2n}}, \ldots, \frac{1}{\sum_{n=1}^{N} H_{Mn}}\right)$$

- $\mathbf{C}$ for column sum normalization:

$$\mathbf{C} = \text{diag}\left(\frac{1}{\sum_{m=1}^{M} H_{m1}}, \frac{1}{\sum_{m=1}^{M} H_{m2}}, \ldots, \frac{1}{\sum_{m=1}^{M} H_{mN}} \right)$$

Appedix: Neumann Series

Neumann series is the operator generalization of the sum of a geometric series:

$$\frac{1}{1-x} = \sum_{j=0}^{\infty} x^j,$$

- x: Scalar
- The series converges uniformly if $|x| < 1$.

Neumann series

$$[\mathbf{I} - \mathbf{\Omega}]^{-1} = \sum_{j=0}^{\infty} \mathbf{\Omega}^j,$$

where

- $\mathbf{I} - \mathbf{\Omega}$: Nonsingular $N \times N$ matrix and $-1 \prec \mathbf{\Omega} \prec 1$

To probve:

$$[\mathbf{I} - \mathbf{\Omega}][\mathbf{I} - \mathbf{\Omega}]^{-1} = [\mathbf{I} - \mathbf{\Omega}] \sum_{j=0}^{\infty} \mathbf{\Omega}^j = \mathbf{I} \sum_{j=0}^{\infty} \mathbf{\Omega}^j - \mathbf{\Omega} \sum_{j=0}^{\infty} \mathbf{\Omega}^j$$

$$= \sum_{j=0}^{\infty} \mathbf{\Omega}^j - \mathbf{\Omega} \sum_{j=0}^{\infty} \mathbf{\Omega}^j = \left[\mathbf{\Omega}^0 + \sum_{j=1}^{\infty} \mathbf{\Omega}^j \right] - \sum_{j=0}^{\infty} \mathbf{\Omega}^{j+1}, \quad m = j - 1$$

$$= \left[\mathbf{I} + \sum_{m=1}^{\infty} \mathbf{\Omega}^{m+1} \right] - \sum_{j=0}^{\infty} \mathbf{\Omega}^{j+1} = \mathbf{I}$$